

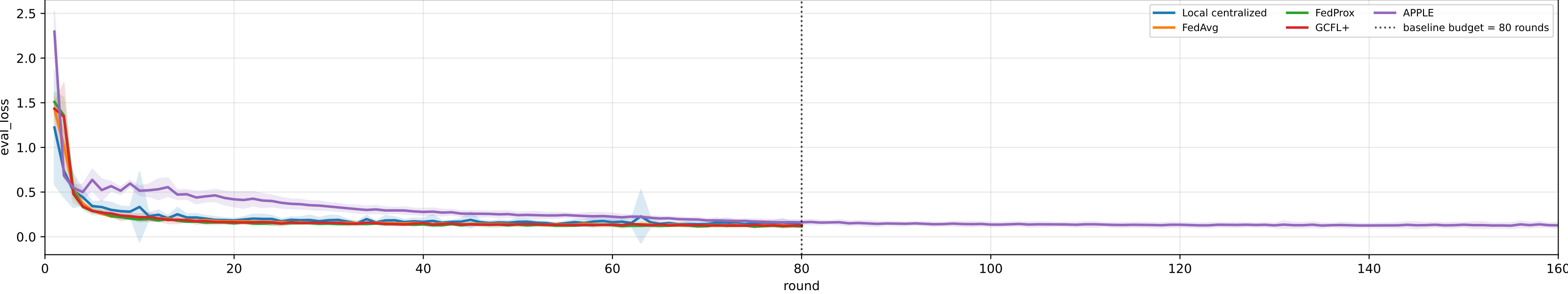
Controlled task-specialization level: Mid | aggregated / pooled performance over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
family=specialized_cycle3 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
family=specialized_cycle4 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.3% | cycle4=visible 6.6%±0.0% / true 6.6% |
family=specialized_cycle5 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
family=specialized_cycle6 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.50 | task_jsd_mean=0.028 | task_jsd_median=0.028 | task_jsd_max=0.029 |
family_centroid_jsd_mean=0.028

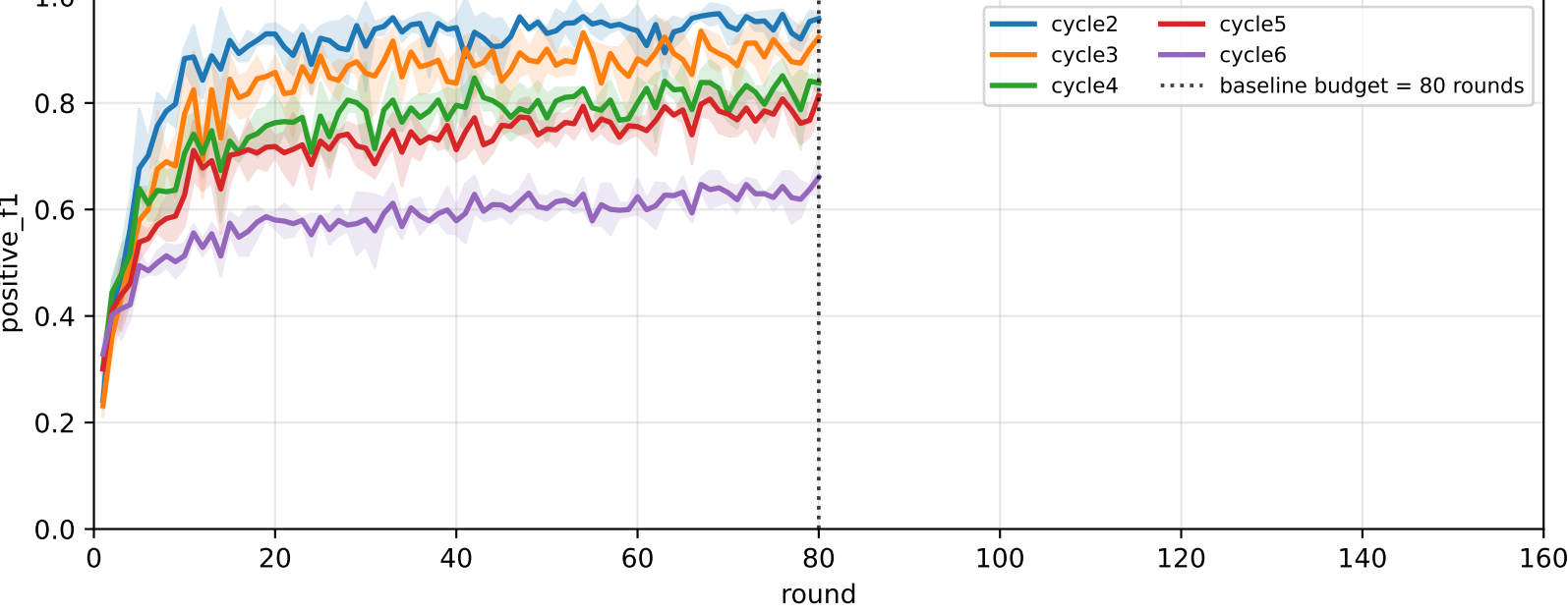
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	81.6% ± 0.7%	84.6% ± 0.7%	96.9% ± 0.3%	95.3% ± 0.9%	87.5% ± 0.4%	77.3% ± 1.9%	65.6% ± 0.6%
FedAvg	78.7% ± 2.2%	81.3% ± 1.4%	94.2% ± 1.3%	92.5% ± 1.6%	79.5% ± 0.1%	74.6% ± 3.1%	65.8% ± 4.6%
FedProx	79.9% ± 1.0%	83.0% ± 0.8%	96.5% ± 1.2%	94.9% ± 0.6%	82.5% ± 1.4%	77.0% ± 2.4%	64.2% ± 3.0%
GCFL+	78.8% ± 0.3%	81.8% ± 0.7%	93.9% ± 2.4%	93.2% ± 1.8%	82.4% ± 2.3%	76.2% ± 0.9%	63.1% ± 0.8%
APPLE	80.1% ± 3.1%	82.9% ± 2.5%	96.0% ± 0.9%	92.4% ± 1.6%	82.0% ± 4.2%	77.6% ± 4.1%	66.0% ± 6.1%

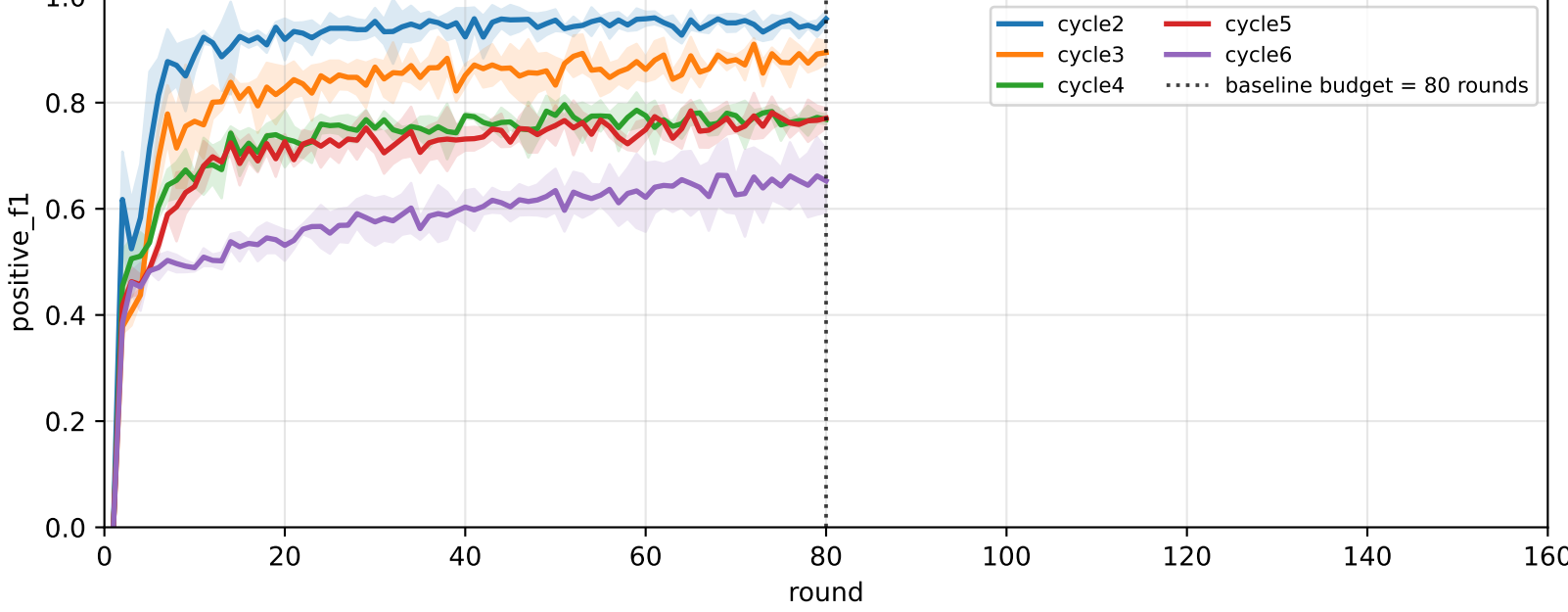
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



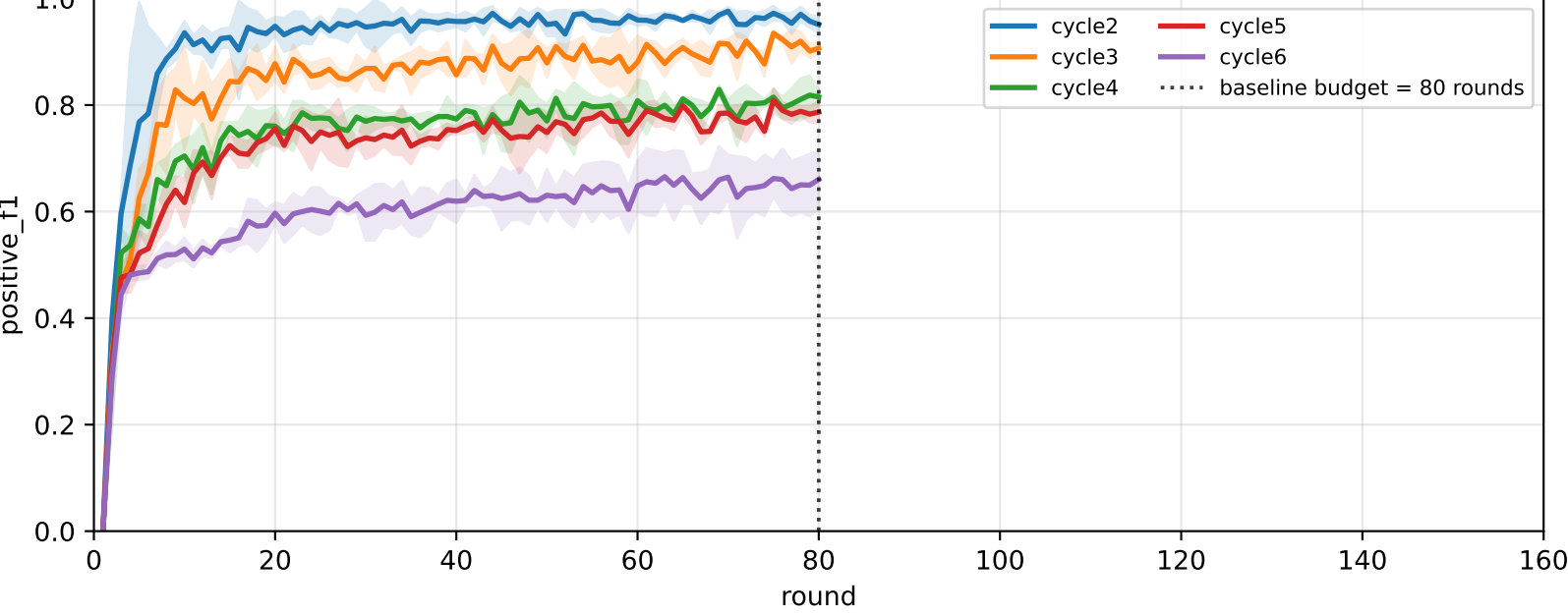
Pooled local centralized positive F1



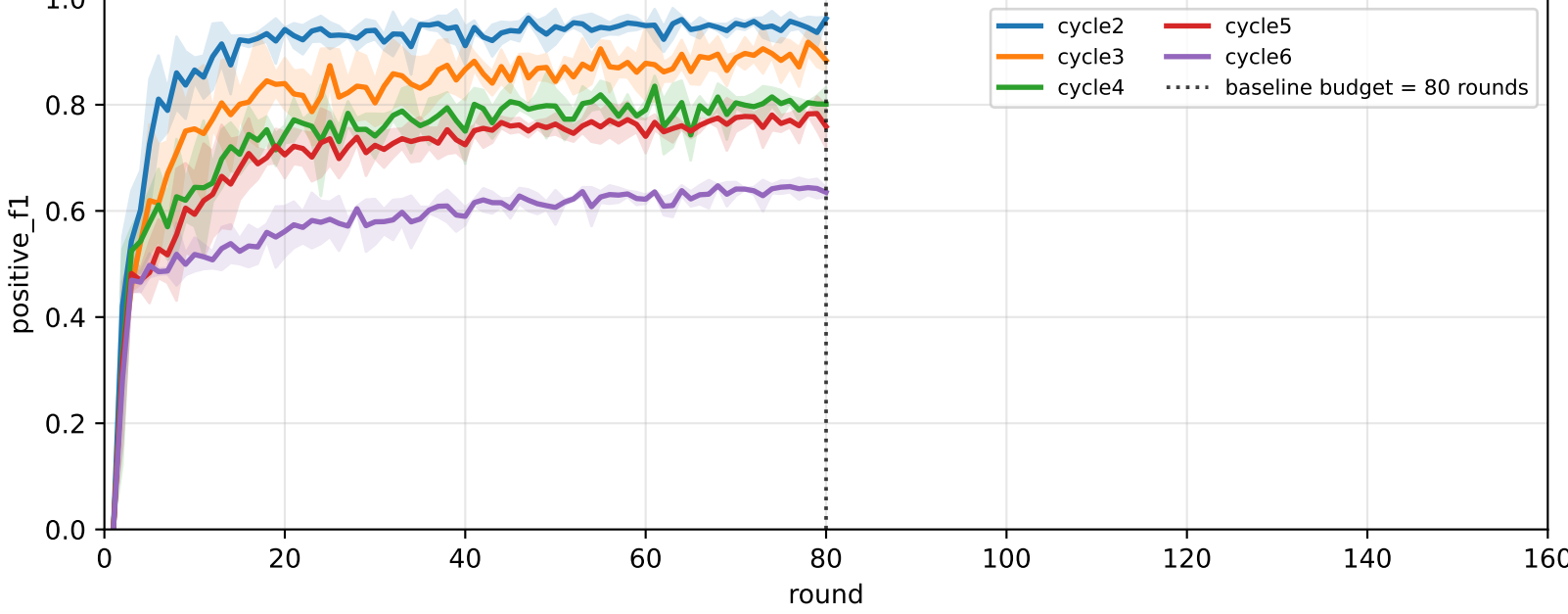
Pooled FedAvg positive F1



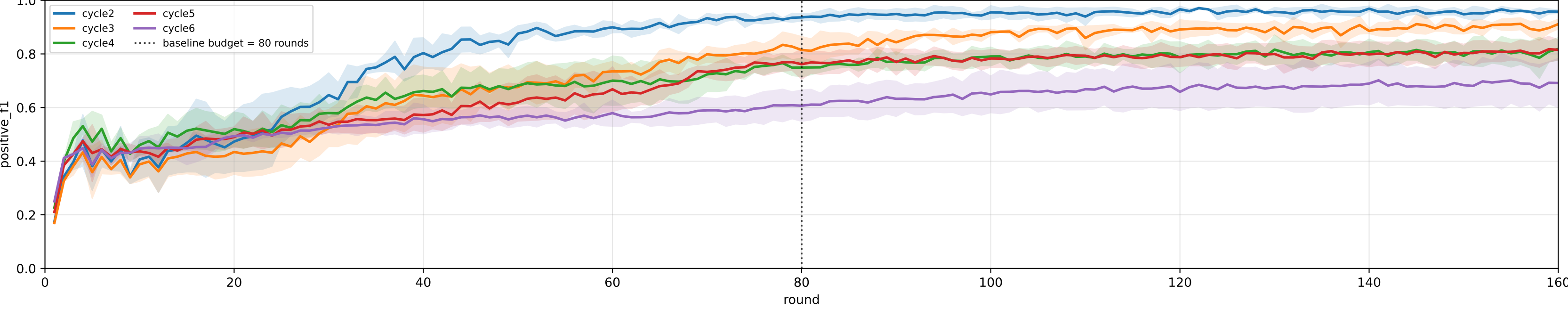
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)



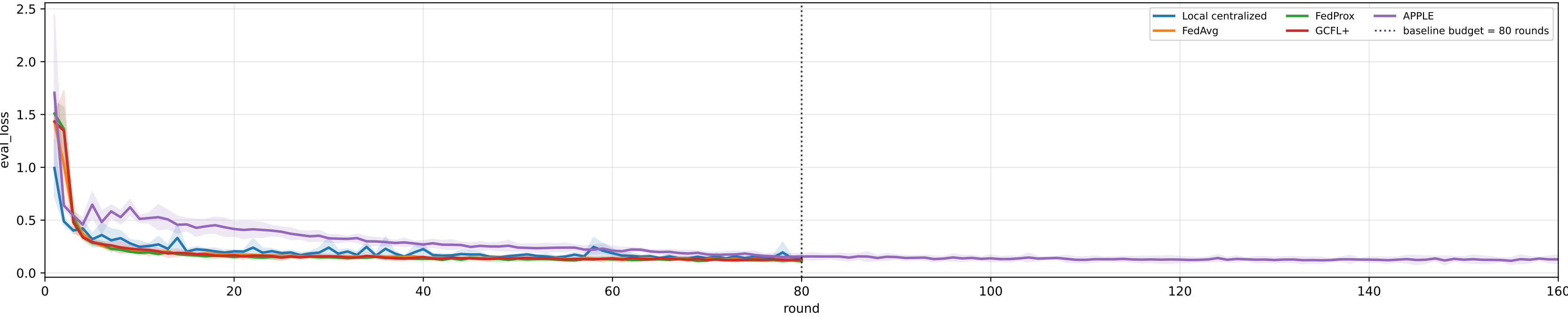
Controlled task-specialization level: Mid | family aggregated / pooled performance |
family=specialized_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% |
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.028 | task_jsd_max=0.029 |
family_centroid_jsd_mean=0.028

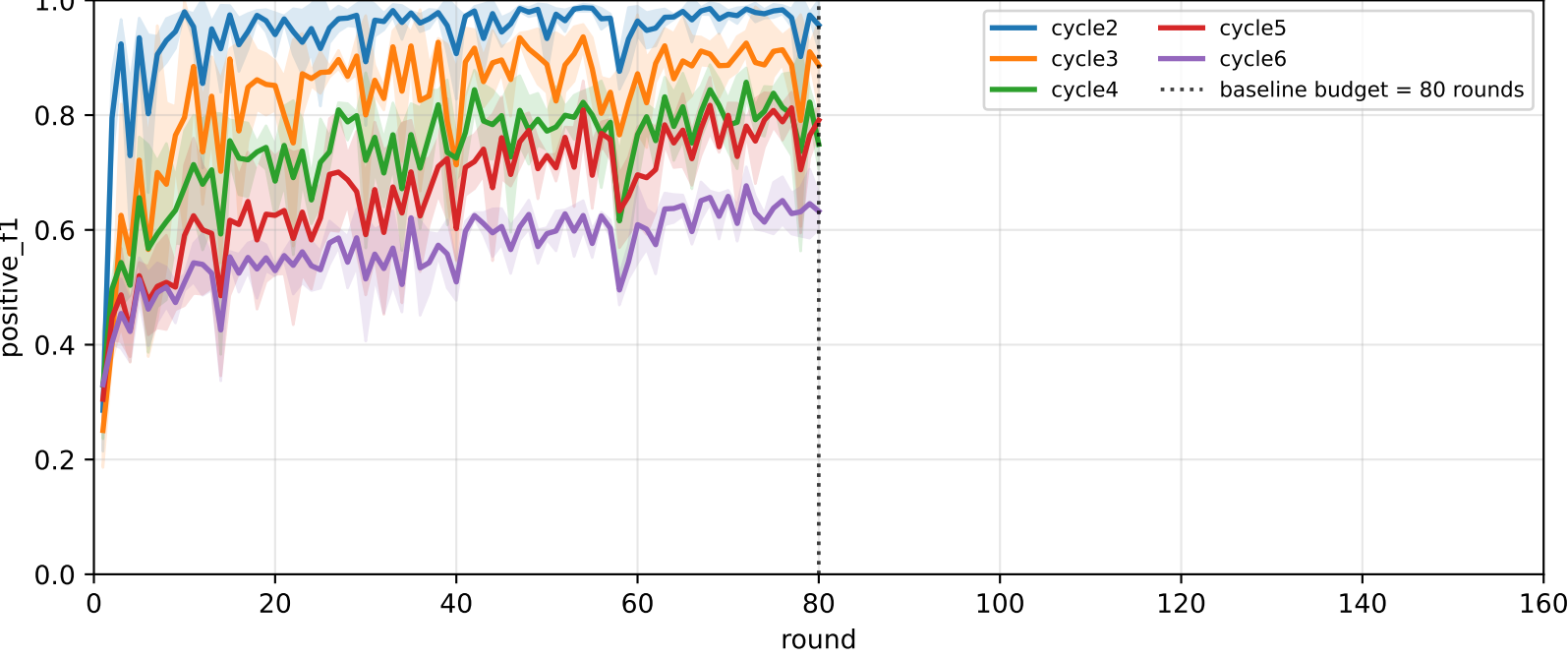
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	81.3% ± 1.7%	84.2% ± 1.3%	98.2% ± 0.7%	94.8% ± 1.7%	86.7% ± 1.3%	75.0% ± 3.4%	66.4% ± 3.0%
FedAvg	78.8% ± 1.9%	81.4% ± 1.1%	94.8% ± 1.6%	92.9% ± 1.0%	79.0% ± 0.6%	74.7% ± 2.7%	65.7% ± 4.6%
FedProx	80.2% ± 0.8%	83.4% ± 0.6%	97.4% ± 0.9%	95.2% ± 0.5%	82.4% ± 2.0%	77.5% ± 1.9%	64.4% ± 2.7%
GCFL+	79.4% ± 0.2%	82.3% ± 0.5%	94.2% ± 2.1%	93.7% ± 1.5%	83.1% ± 2.1%	77.2% ± 1.0%	63.3% ± 0.8%
APPLE	80.6% ± 2.9%	83.6% ± 2.6%	98.0% ± 0.3%	93.4% ± 2.5%	83.4% ± 6.1%	78.0% ± 4.0%	65.2% ± 3.7%

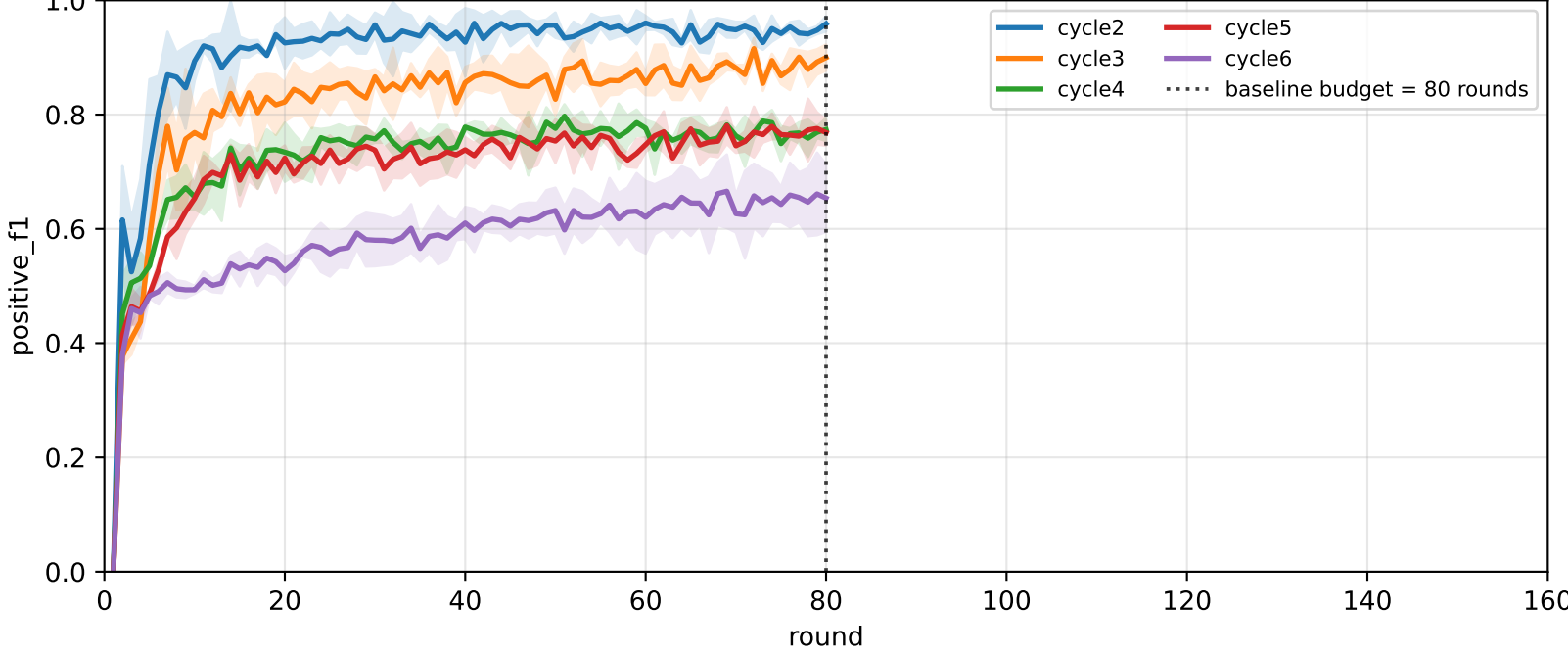
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



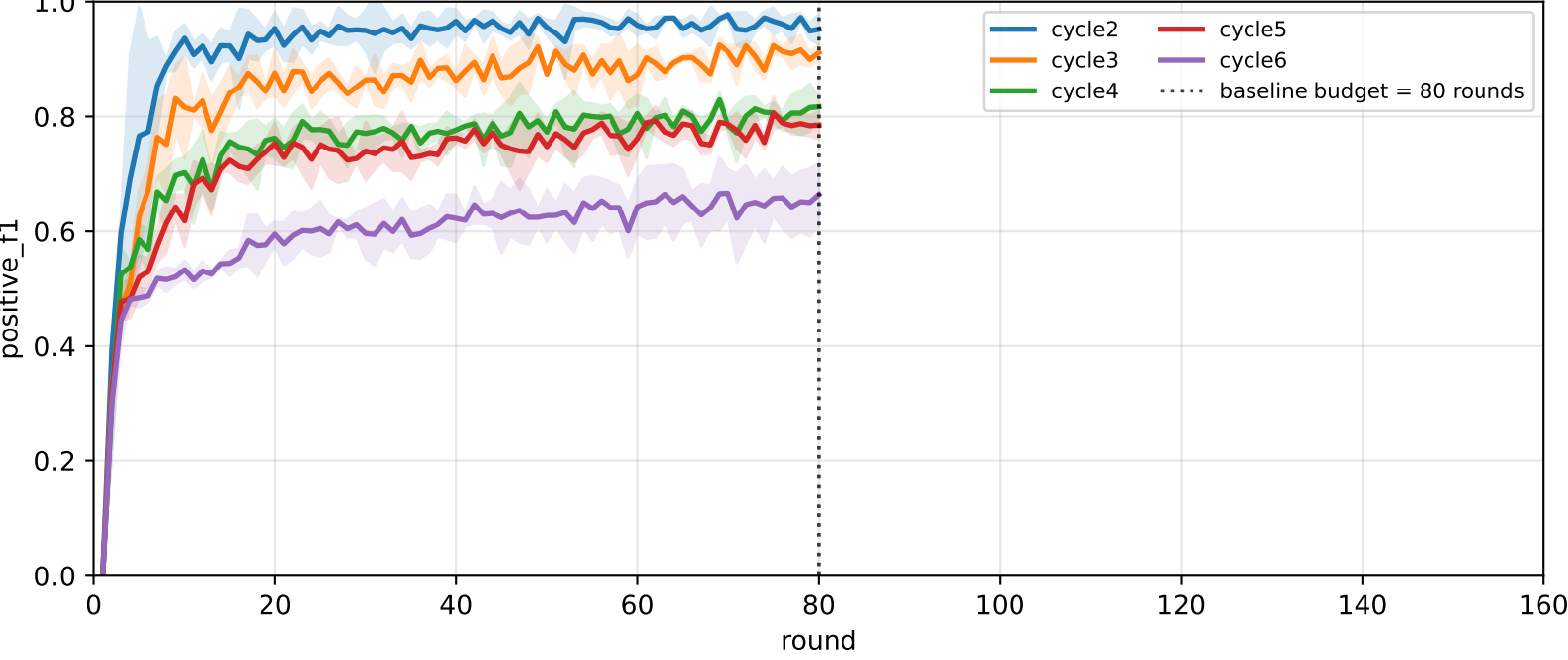
Pooled local centralized positive F1



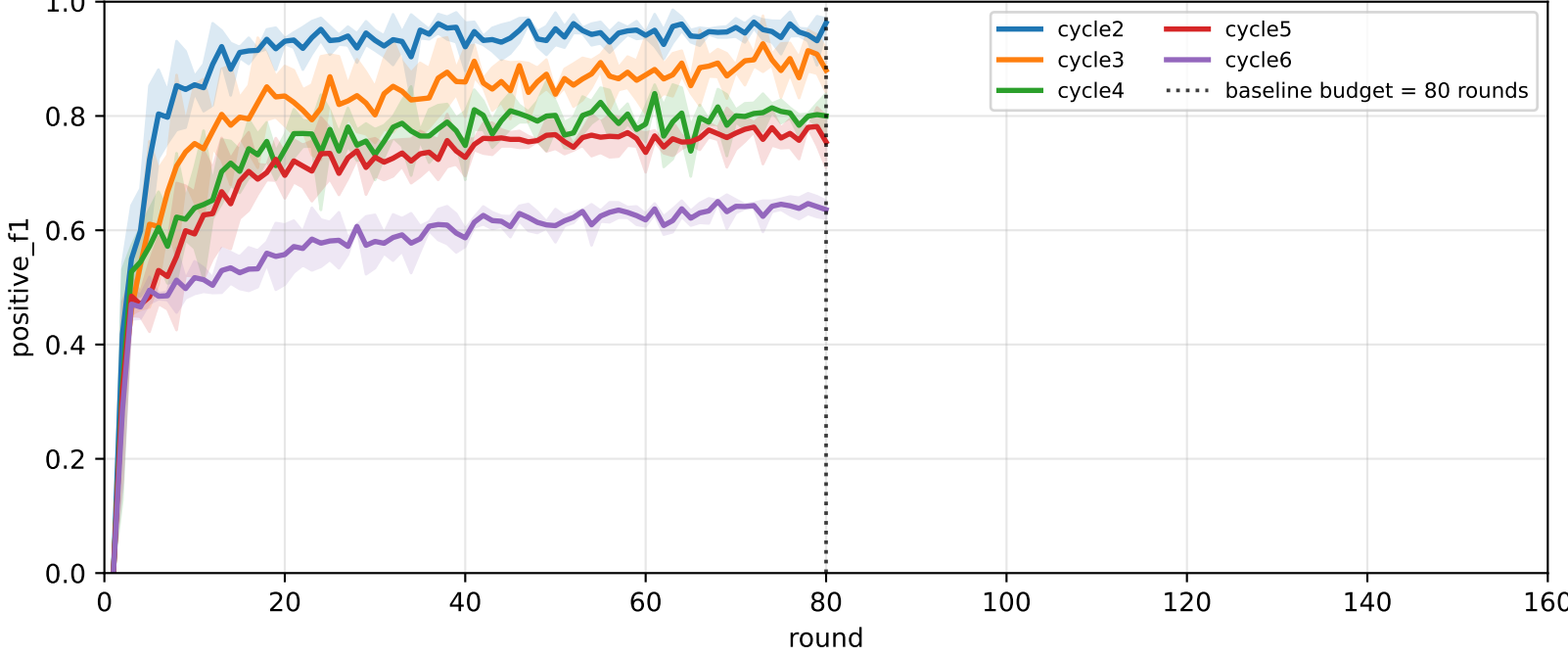
Pooled FedAvg positive F1



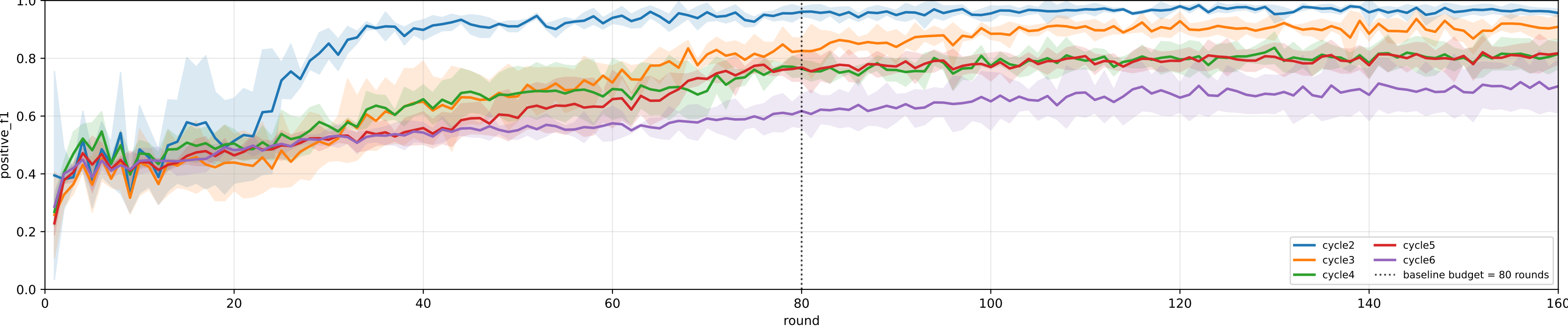
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)



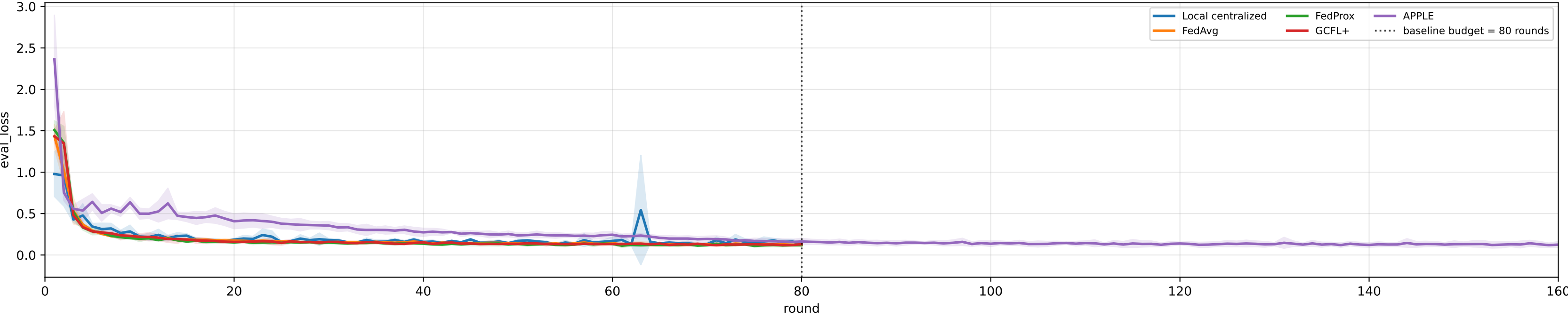
Controlled task-specialization level: Mid | family aggregated / pooled performance |
family=specialized_cycle3 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle3 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% |
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.028 | task_jsd_max=0.029 |
family_centroid_jsd_mean=0.028

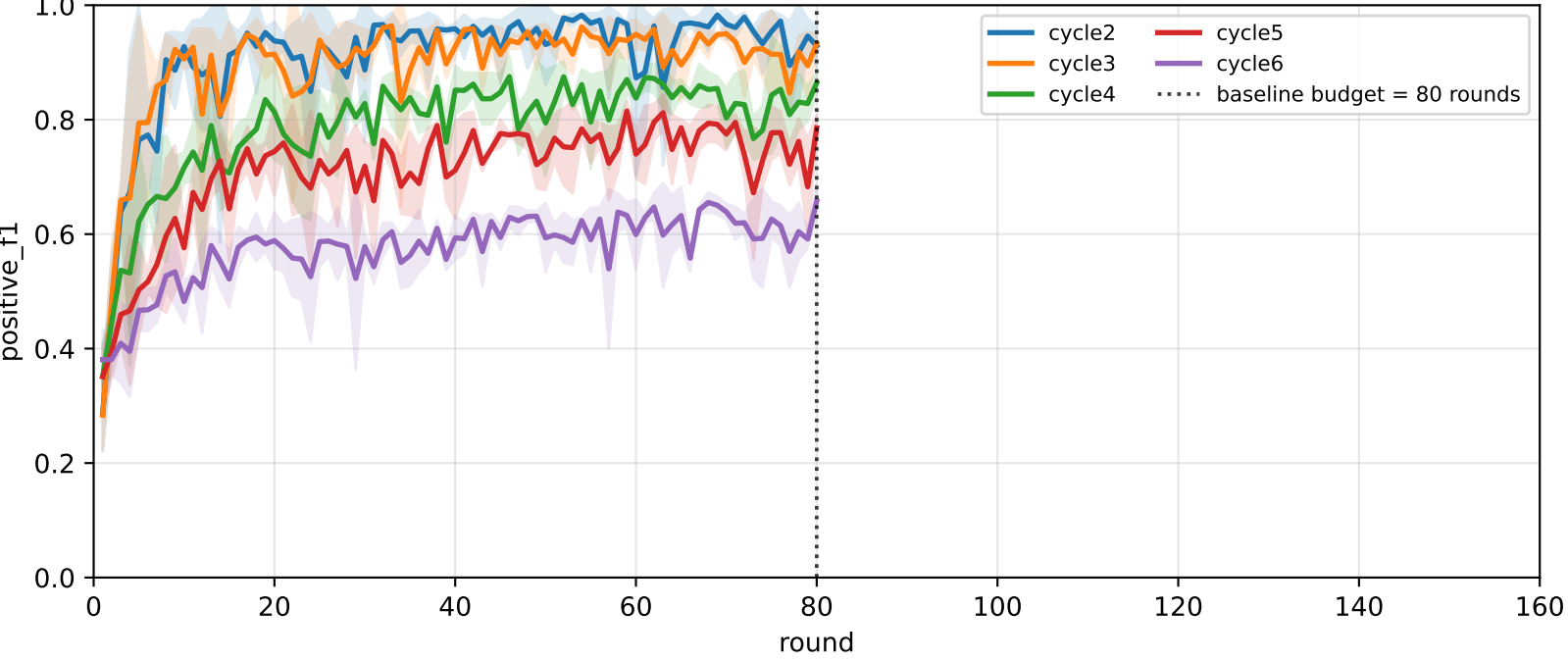
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	81.1% ± 0.2%	84.7% ± 0.1%	97.7% ± 0.8%	96.9% ± 0.8%	90.8% ± 0.3%	74.1% ± 1.7%	64.0% ± 0.4%
FedAvg	78.7% ± 2.1%	81.3% ± 1.4%	93.5% ± 0.9%	92.5% ± 1.6%	80.1% ± 0.7%	74.1% ± 2.8%	65.9% ± 4.7%
FedProx	79.9% ± 0.8%	83.0% ± 0.8%	96.7% ± 1.0%	95.1% ± 0.9%	82.6% ± 1.7%	76.5% ± 2.9%	64.3% ± 3.0%
GCFL+	79.0% ± 0.6%	81.9% ± 0.8%	93.4% ± 1.8%	92.9% ± 2.0%	83.4% ± 2.4%	76.8% ± 0.4%	63.1% ± 0.9%
APPLE	79.4% ± 3.6%	82.6% ± 2.4%	96.2% ± 0.5%	94.2% ± 2.2%	80.3% ± 2.8%	77.9% ± 4.2%	64.2% ± 7.9%

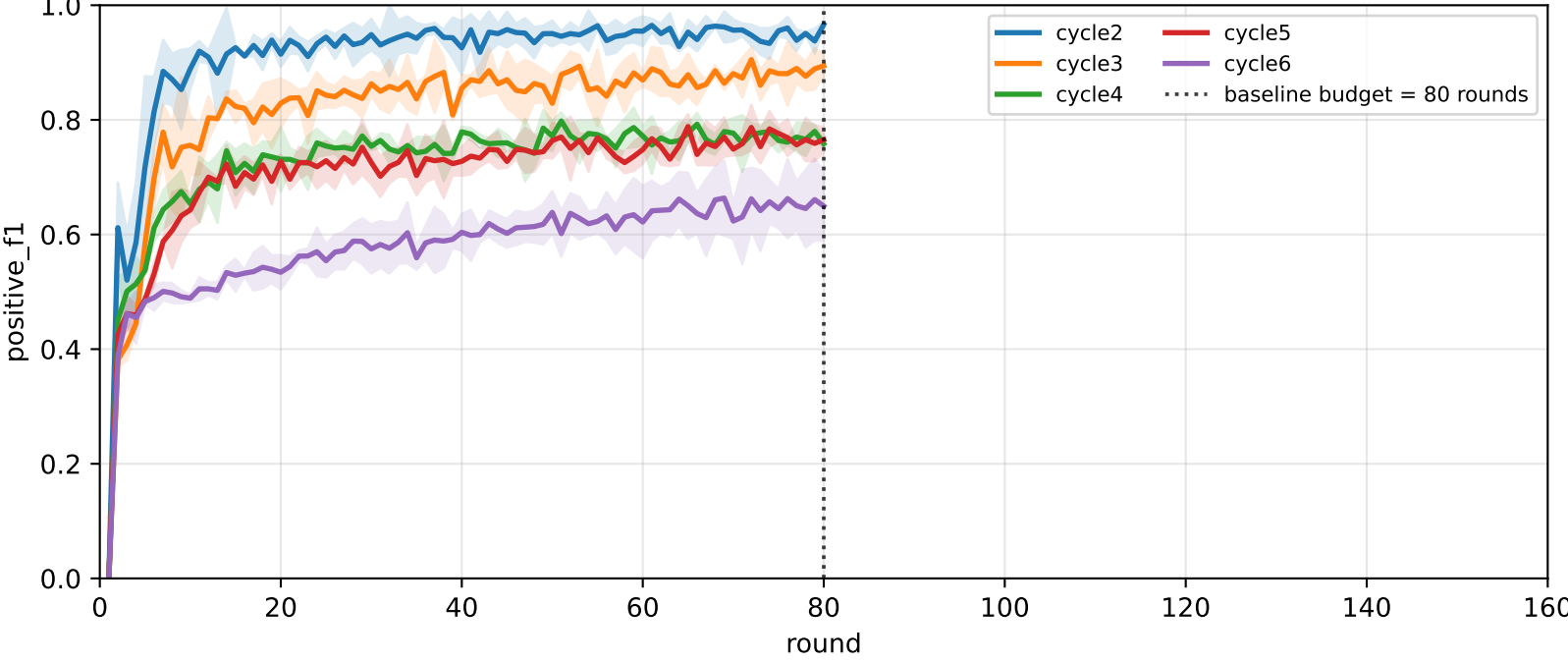
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



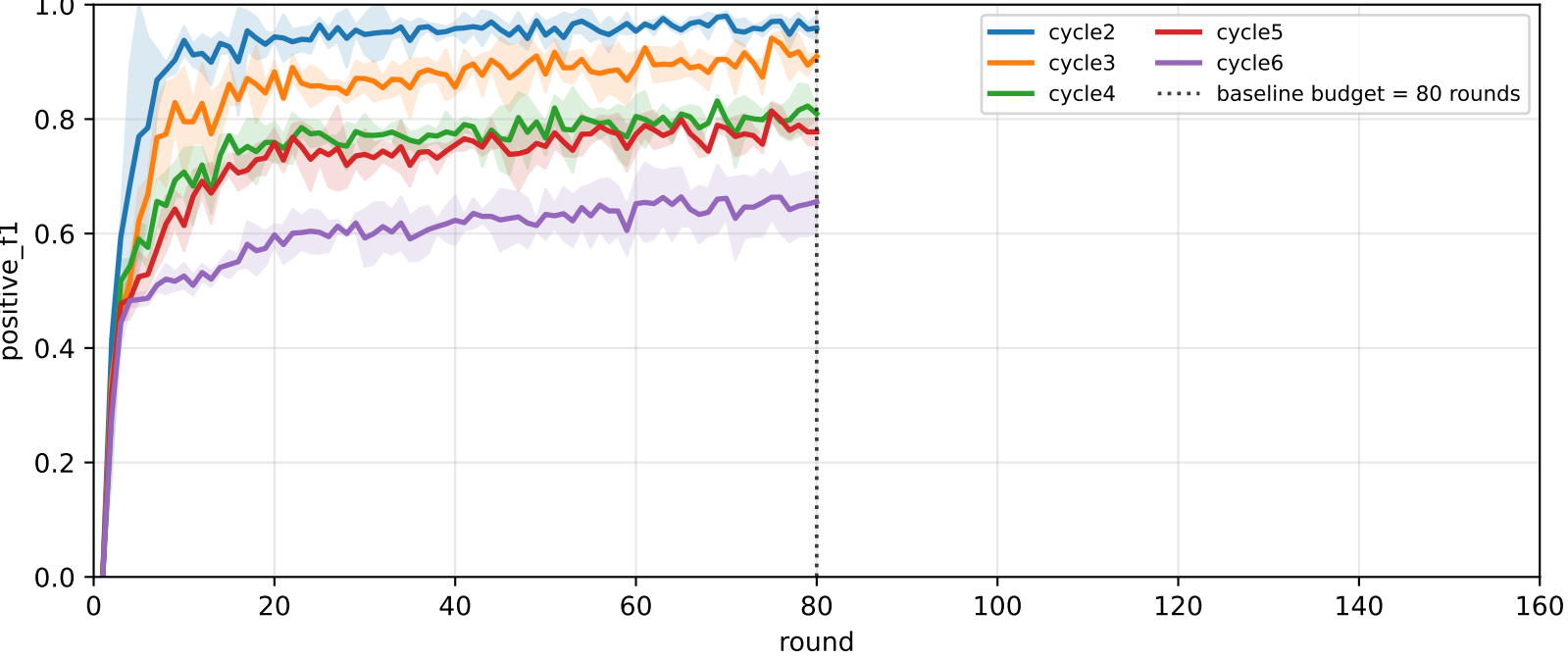
Pooled local centralized positive F1



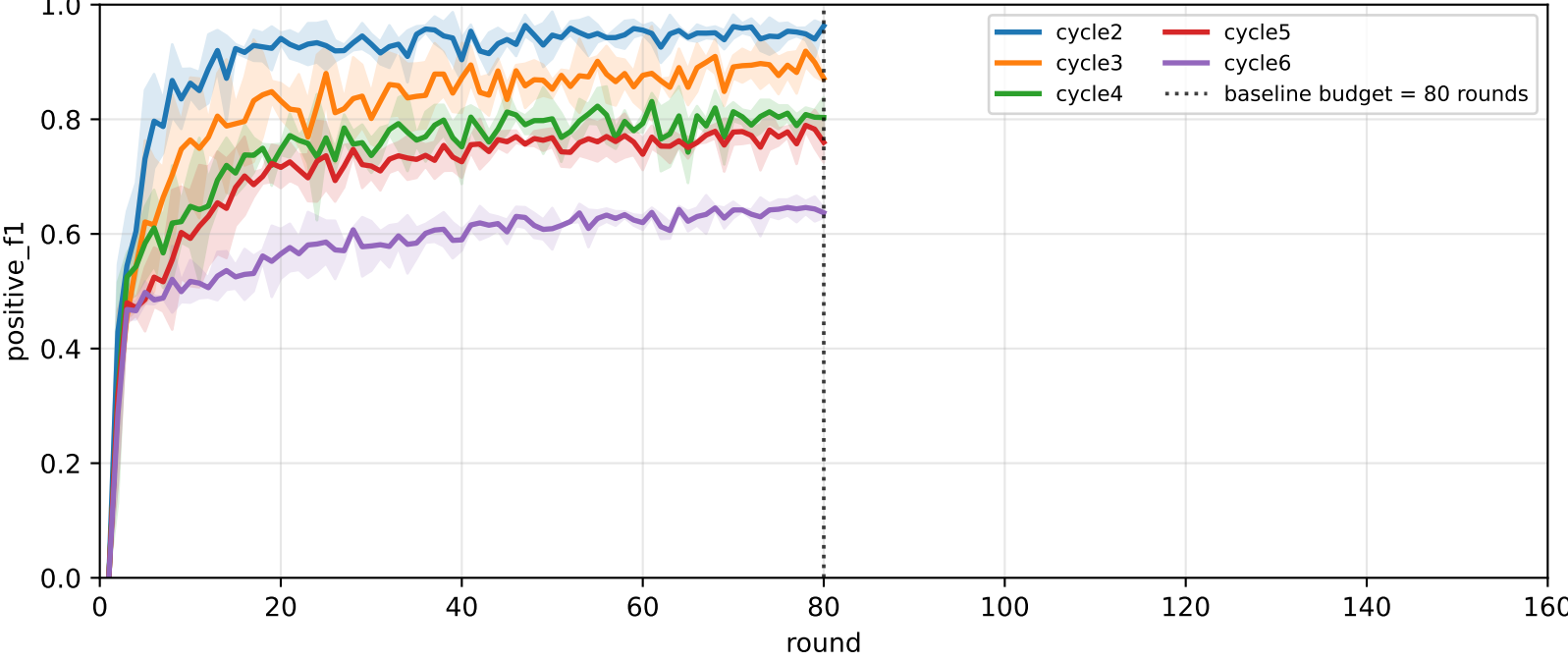
Pooled FedAvg positive F1



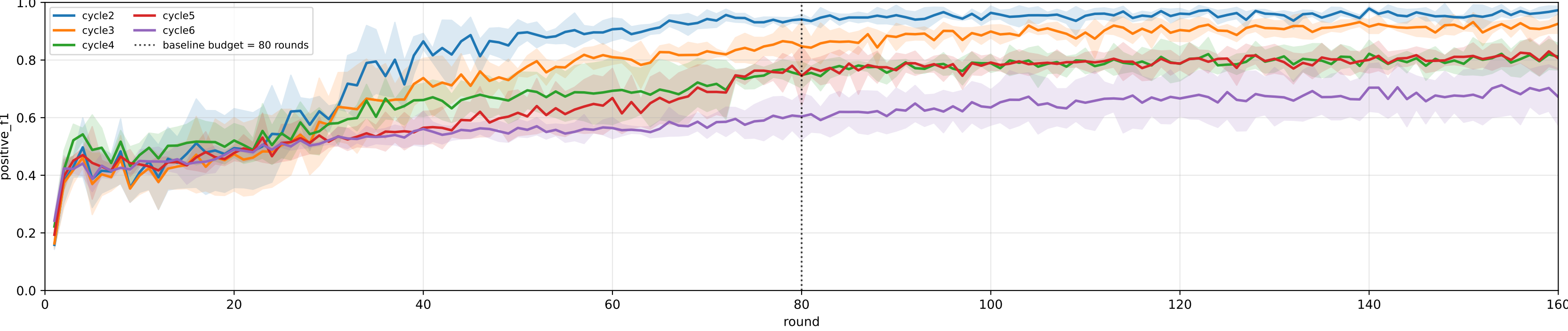
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)



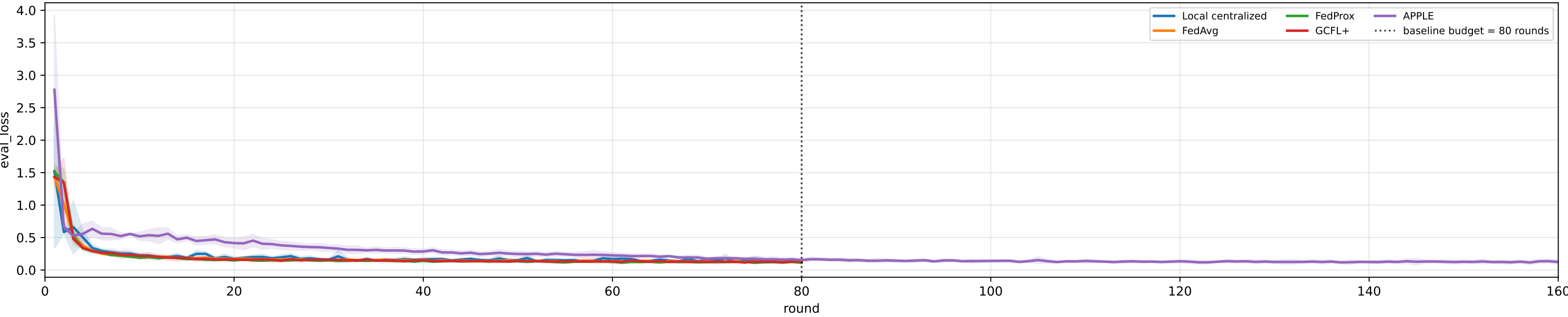
Controlled task-specialization level: Mid | family aggregated / pooled performance |
family=specialized_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle4 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% |
cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% |
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.028 | task_jsd_max=0.029 |
family_centroid_jsd_mean=0.028

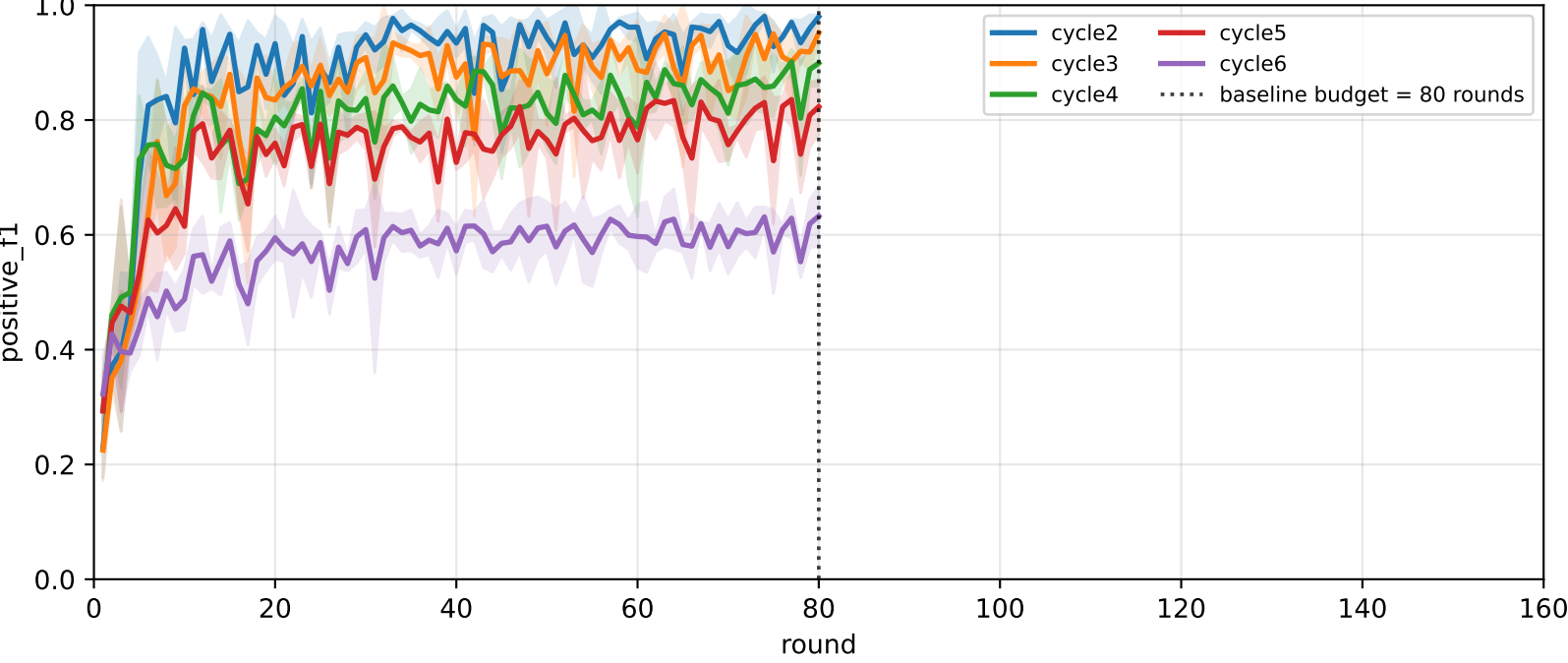
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	82.3% ± 1.4%	85.7% ± 1.2%	97.7% ± 1.1%	96.2% ± 0.1%	89.8% ± 1.6%	80.9% ± 4.0%	63.8% ± 1.5%
FedAvg	78.7% ± 2.2%	81.3% ± 1.3%	94.8% ± 2.3%	91.9% ± 1.2%	79.7% ± 0.4%	74.3% ± 3.1%	65.7% ± 4.8%
FedProx	79.6% ± 1.2%	82.7% ± 0.9%	96.4% ± 1.6%	94.1% ± 0.6%	82.2% ± 1.5%	76.9% ± 2.8%	63.9% ± 2.7%
GCFL+	78.5% ± 0.6%	81.3% ± 0.9%	93.4% ± 3.1%	92.7% ± 2.8%	81.8% ± 2.6%	75.3% ± 0.9%	63.3% ± 0.8%
APPLE	80.8% ± 3.4%	83.4% ± 2.9%	95.9% ± 1.0%	92.5% ± 2.4%	84.3% ± 5.9%	77.4% ± 4.6%	67.1% ± 6.0%

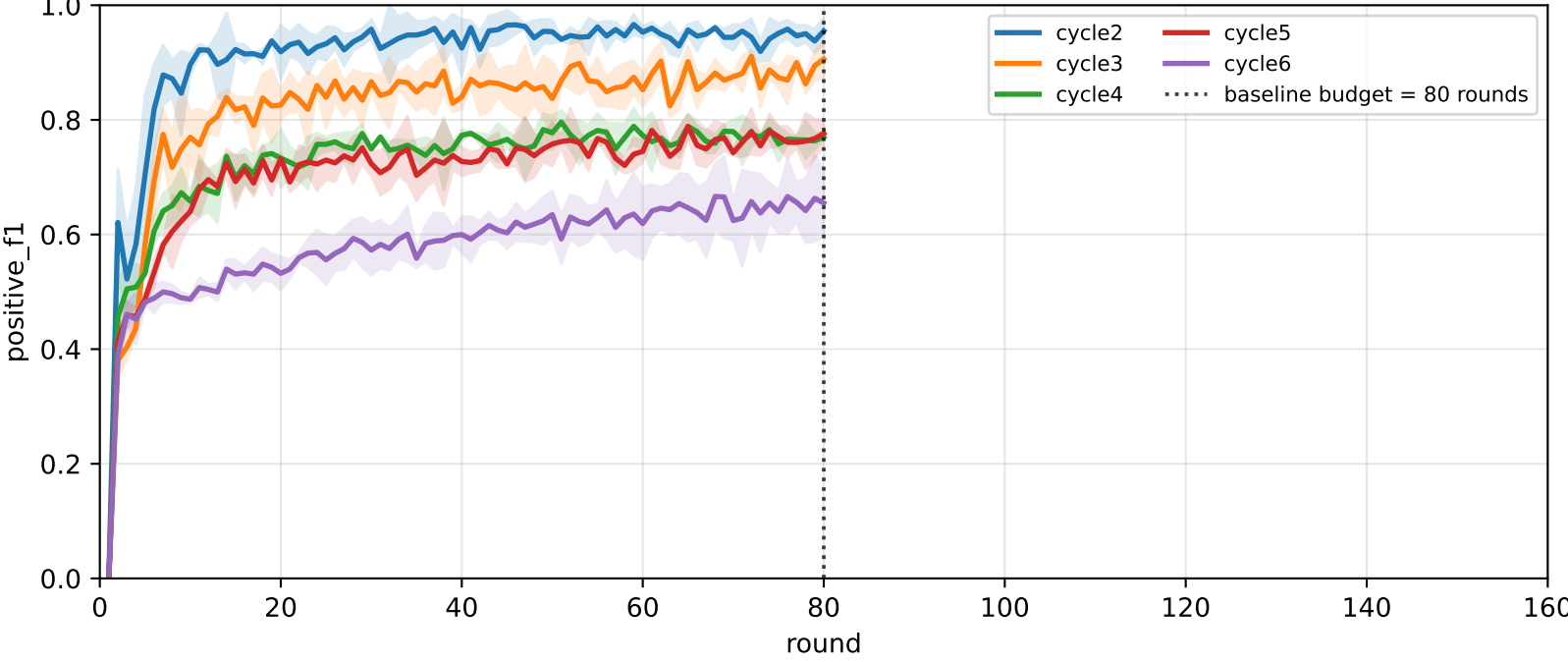
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



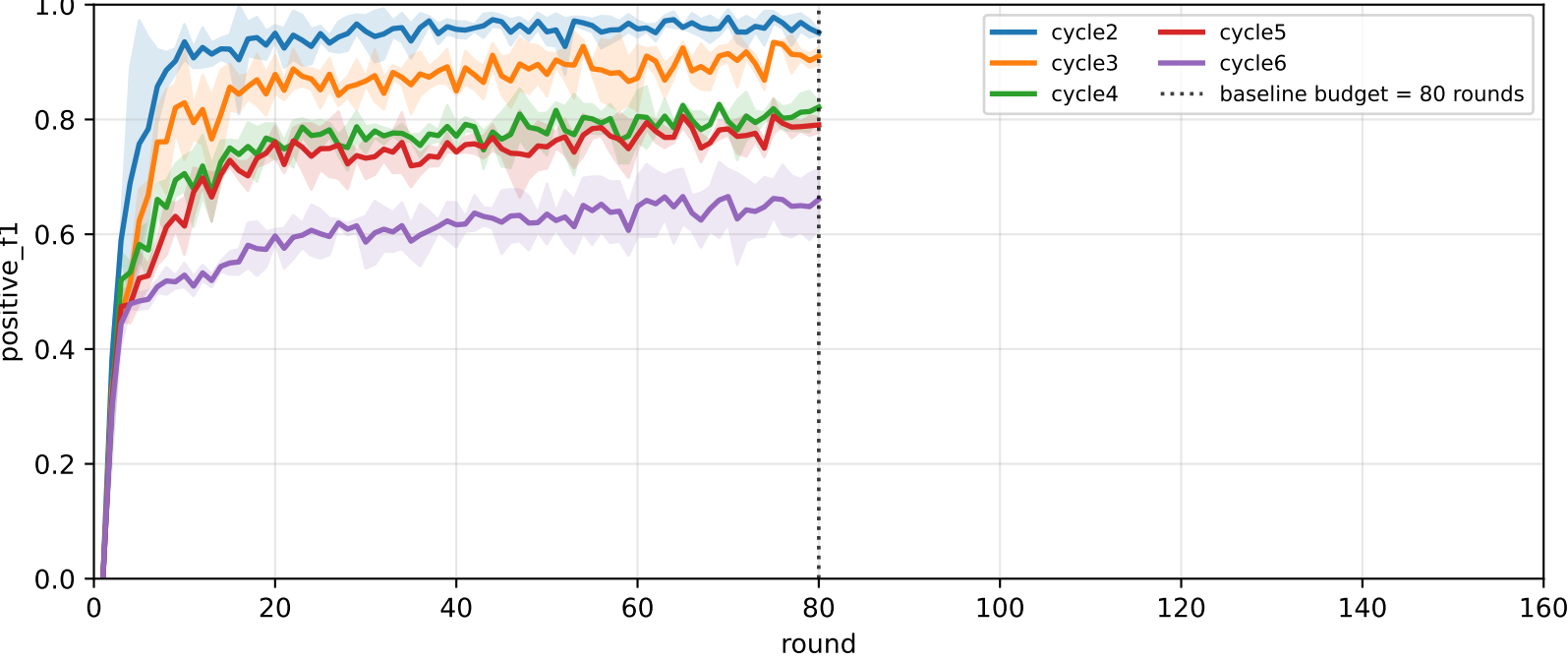
Pooled local centralized positive F1



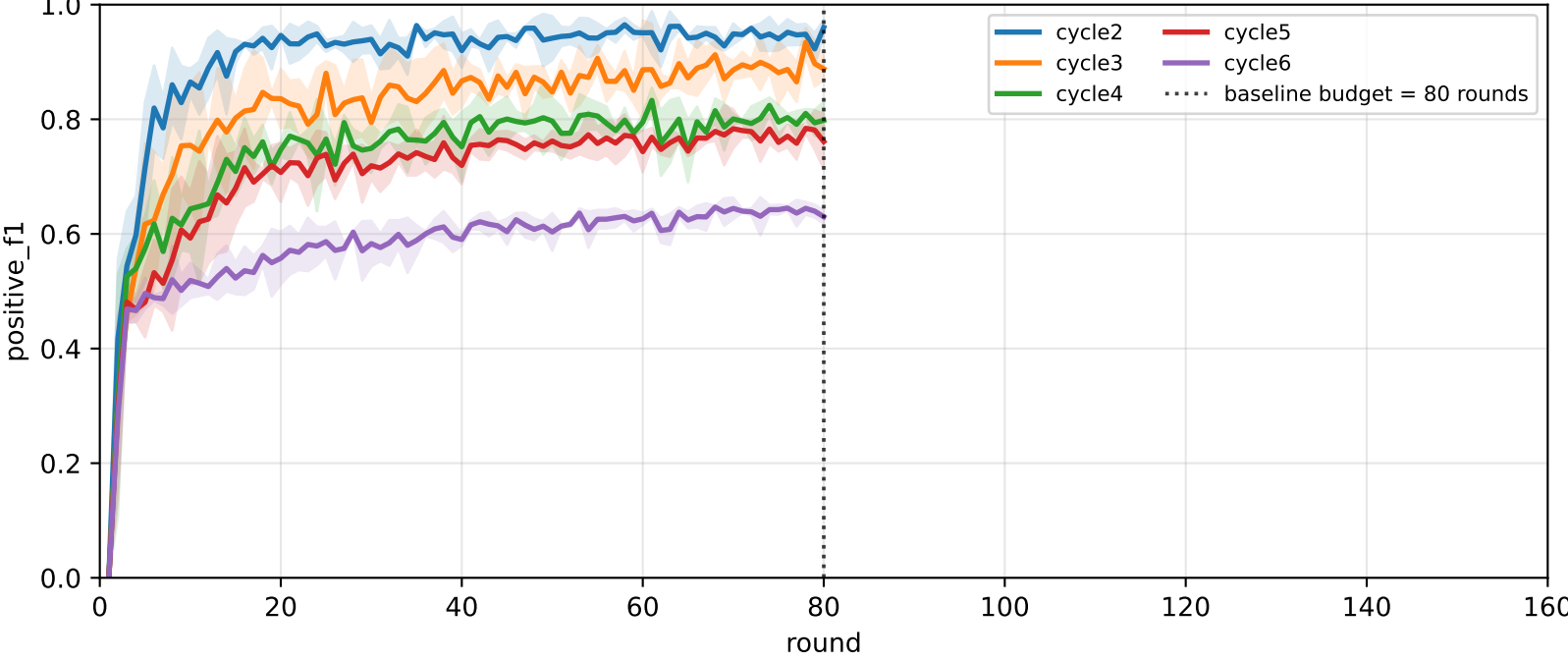
Pooled FedAvg positive F1



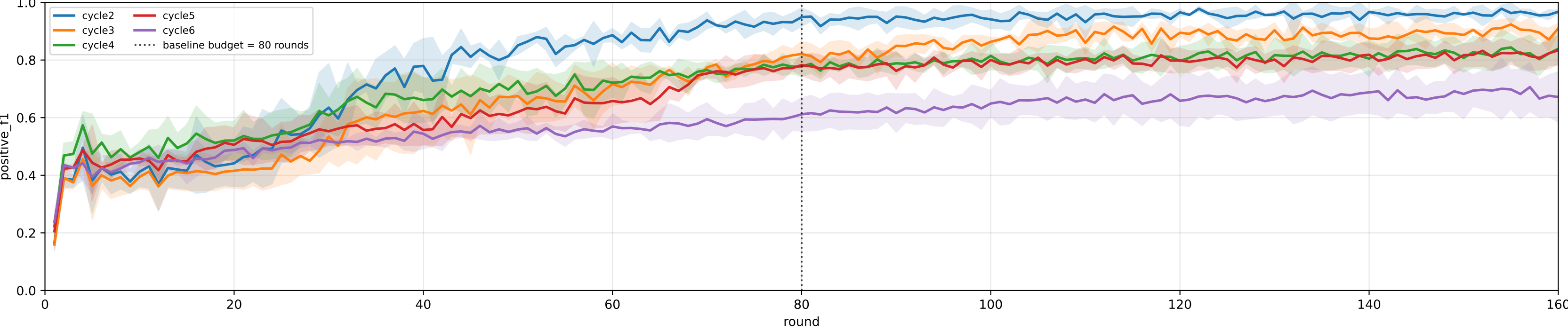
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)



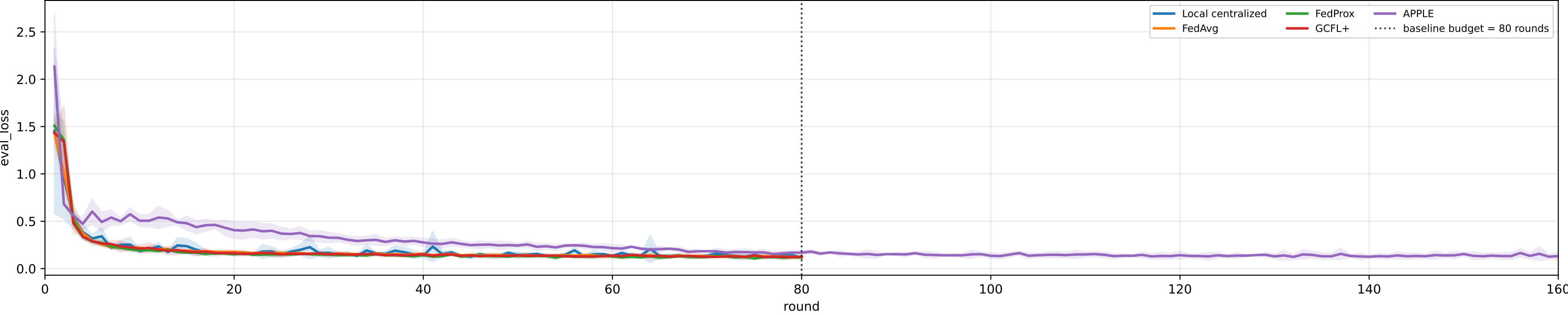
Controlled task-specialization level: Mid | family aggregated / pooled performance |
family=specialized_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle5 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3% |
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.028 | task_jsd_max=0.029 |
family_centroid_jsd_mean=0.028

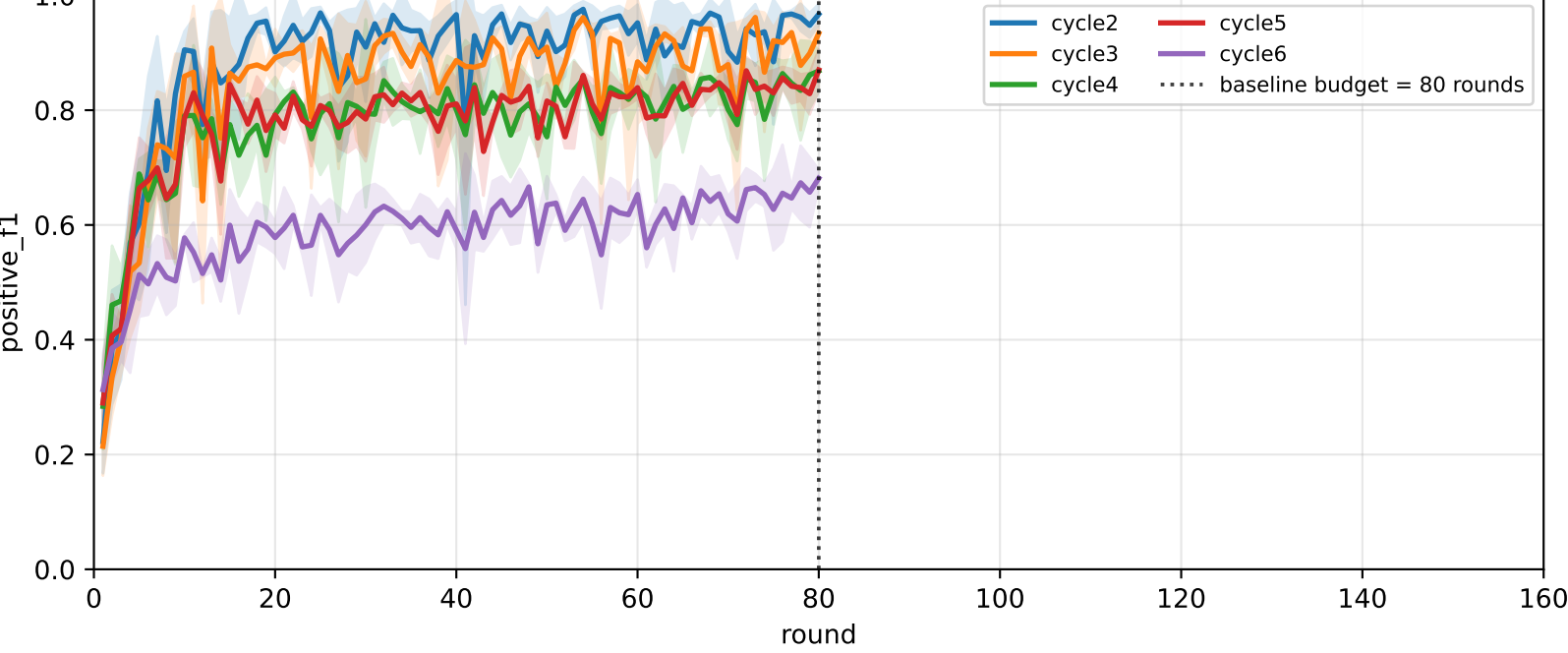
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	82.2% ± 0.8%	84.9% ± 0.9%	95.7% ± 1.8%	95.1% ± 0.8%	86.8% ± 3.3%	80.6% ± 1.1%	66.1% ± 1.3%
FedAvg	78.2% ± 2.1%	80.7% ± 1.3%	93.4% ± 1.5%	91.9% ± 1.8%	78.5% ± 0.4%	74.5% ± 3.9%	65.4% ± 4.6%
FedProx	79.7% ± 0.9%	82.6% ± 0.8%	94.8% ± 2.2%	94.7% ± 1.0%	82.0% ± 1.3%	77.0% ± 2.4%	64.6% ± 3.4%
GCFL+	78.7% ± 0.5%	81.7% ± 0.8%	94.6% ± 1.8%	93.4% ± 1.7%	81.4% ± 2.8%	75.8% ± 1.2%	63.1% ± 0.9%
APPLE	79.1% ± 3.0%	81.9% ± 2.6%	94.9% ± 1.9%	90.8% ± 1.6%	81.7% ± 3.3%	77.9% ± 4.1%	64.1% ± 5.2%

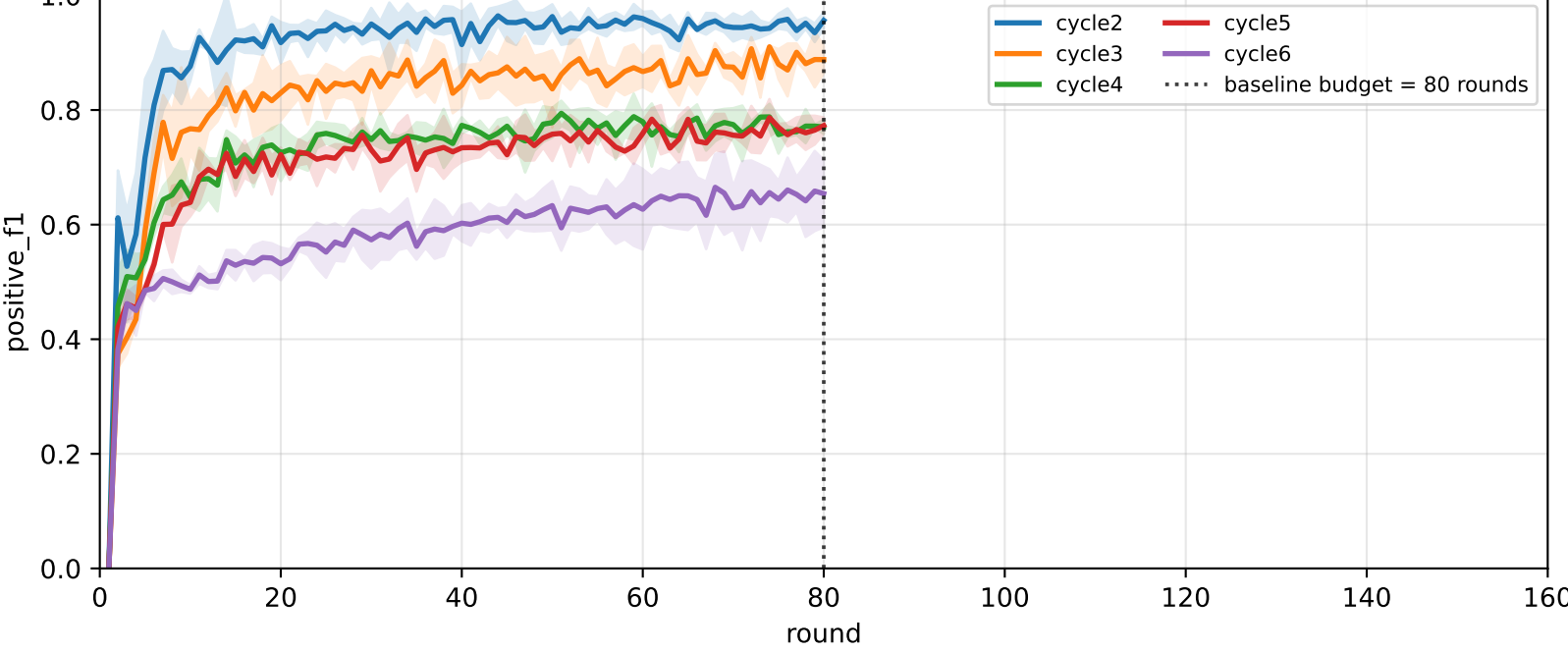
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



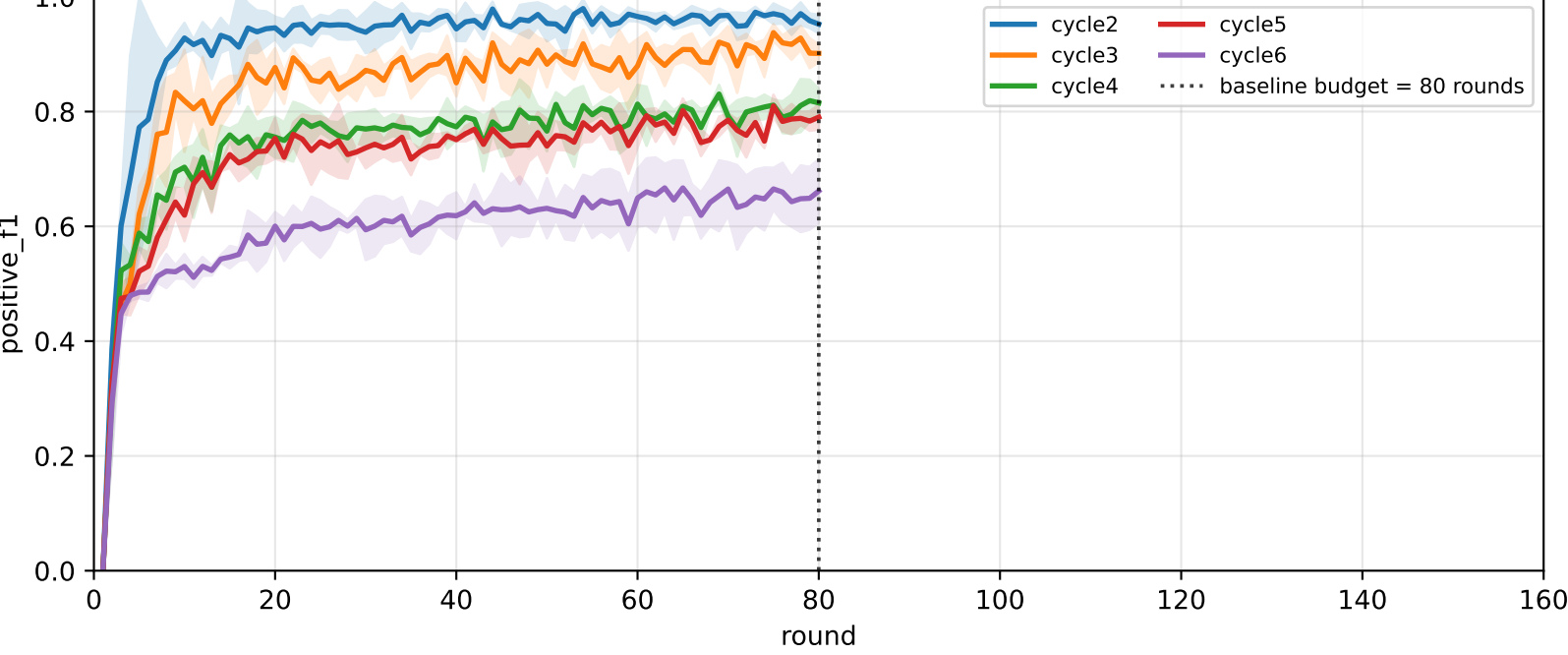
Pooled local centralized positive F1



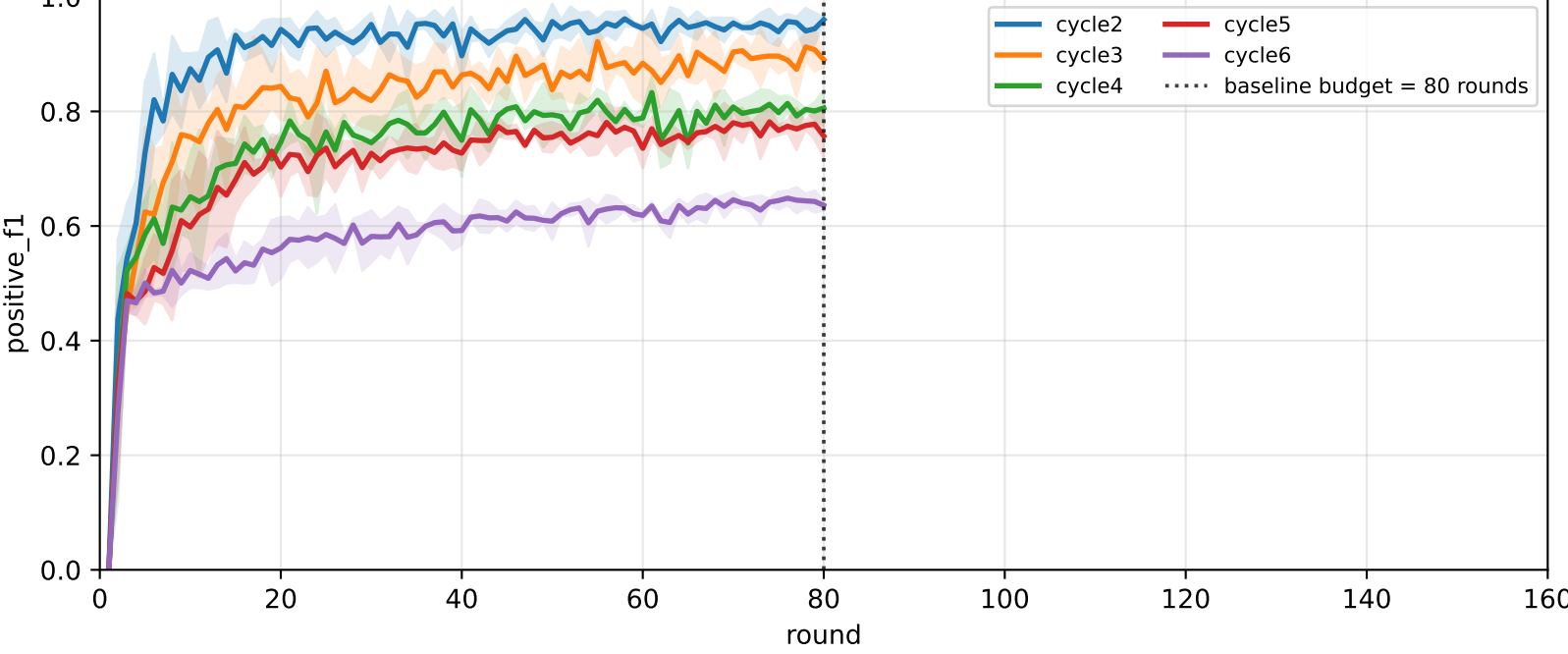
Pooled FedAvg positive F1



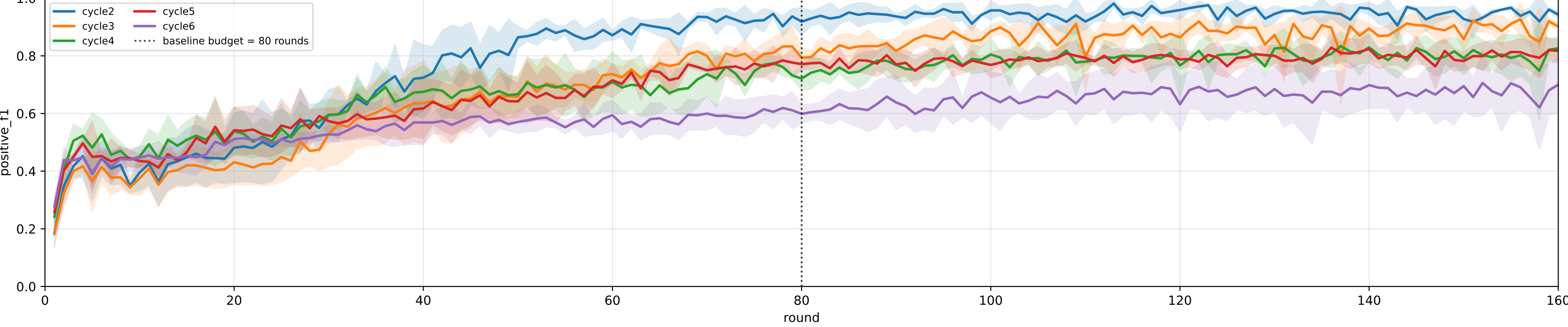
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)



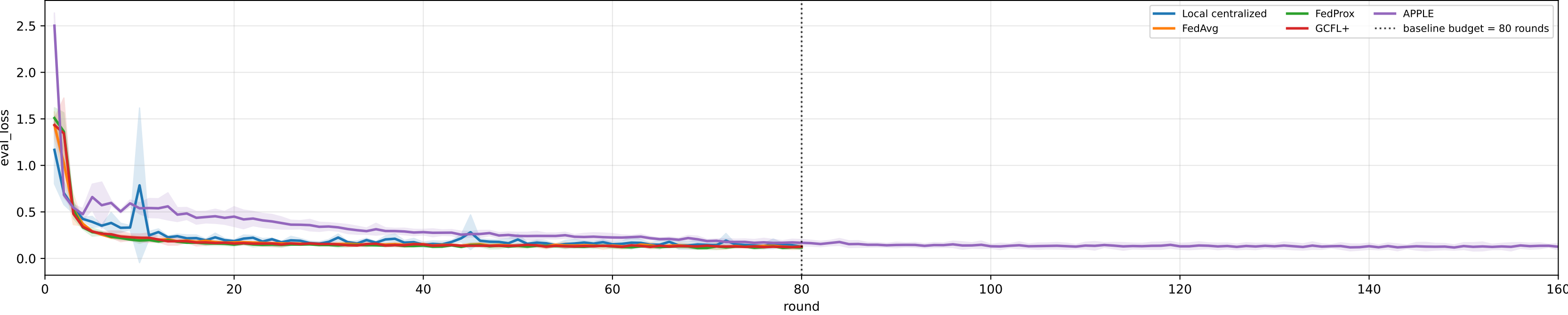
Controlled task-specialization level: Mid | family aggregated / pooled performance |
family=specialized_cycle6 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle6 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%
specialization=mid | specialization_fraction=0.50 | global_visible_support=0.60 | task_jsd_mean=0.028 | task_jsd_median=0.028 | task_jsd_max=0.029 |
family_centroid_jsd_mean=0.028

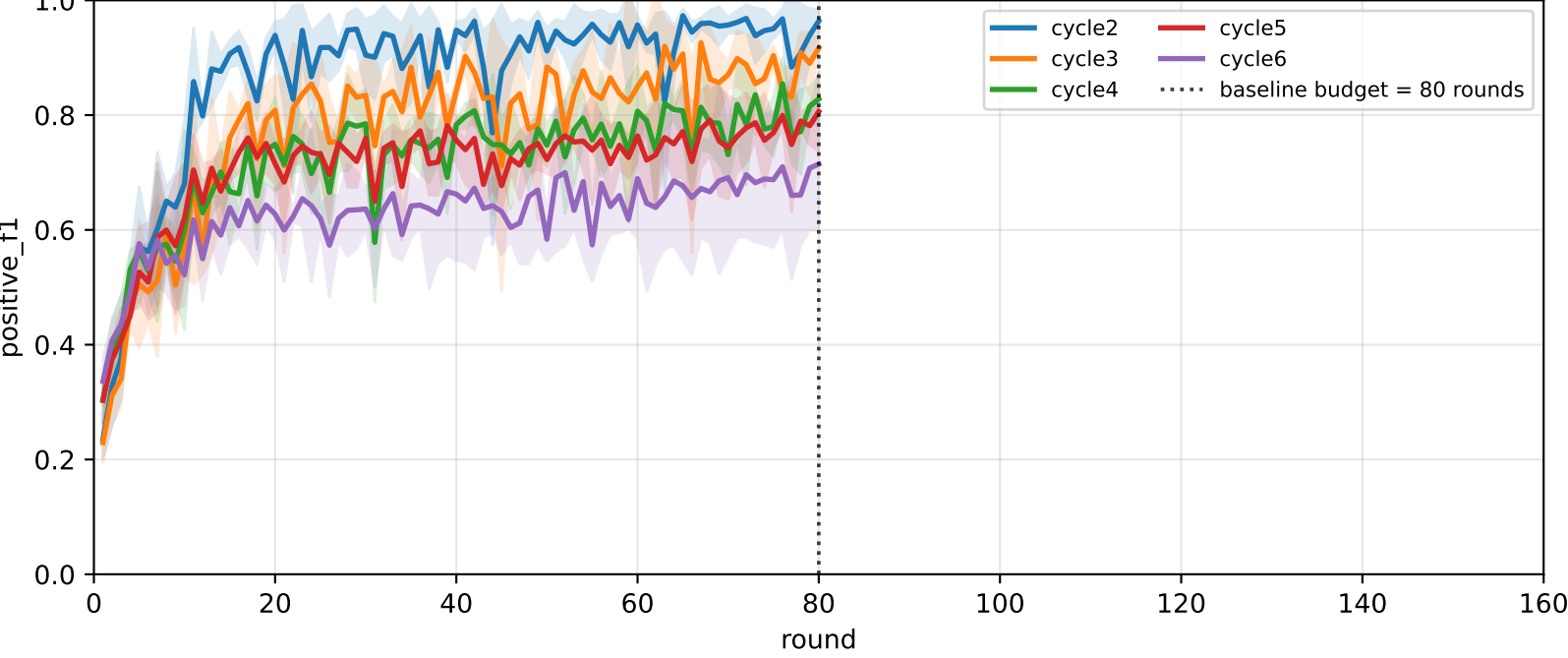
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	81.3% ± 2.7%	83.7% ± 2.0%	95.6% ± 1.4%	93.8% ± 1.7%	84.2% ± 3.5%	76.6% ± 4.7%	68.5% ± 6.6%
FedAvg	79.3% ± 2.6%	81.9% ± 1.9%	94.6% ± 1.4%	93.3% ± 2.5%	80.0% ± 1.1%	75.2% ± 3.1%	66.1% ± 4.7%
FedProx	80.2% ± 1.2%	83.4% ± 0.9%	97.1% ± 1.0%	95.5% ± 0.8%	83.4% ± 1.3%	76.9% ± 2.3%	64.0% ± 3.5%
GCFL+	78.7% ± 0.2%	81.6% ± 0.6%	93.8% ± 3.5%	93.4% ± 1.3%	82.2% ± 2.2%	75.8% ± 1.5%	63.0% ± 1.1%
APPLE	80.6% ± 3.2%	82.8% ± 2.4%	95.0% ± 1.8%	91.5% ± 0.3%	80.7% ± 3.4%	77.1% ± 3.9%	69.9% ± 8.2%

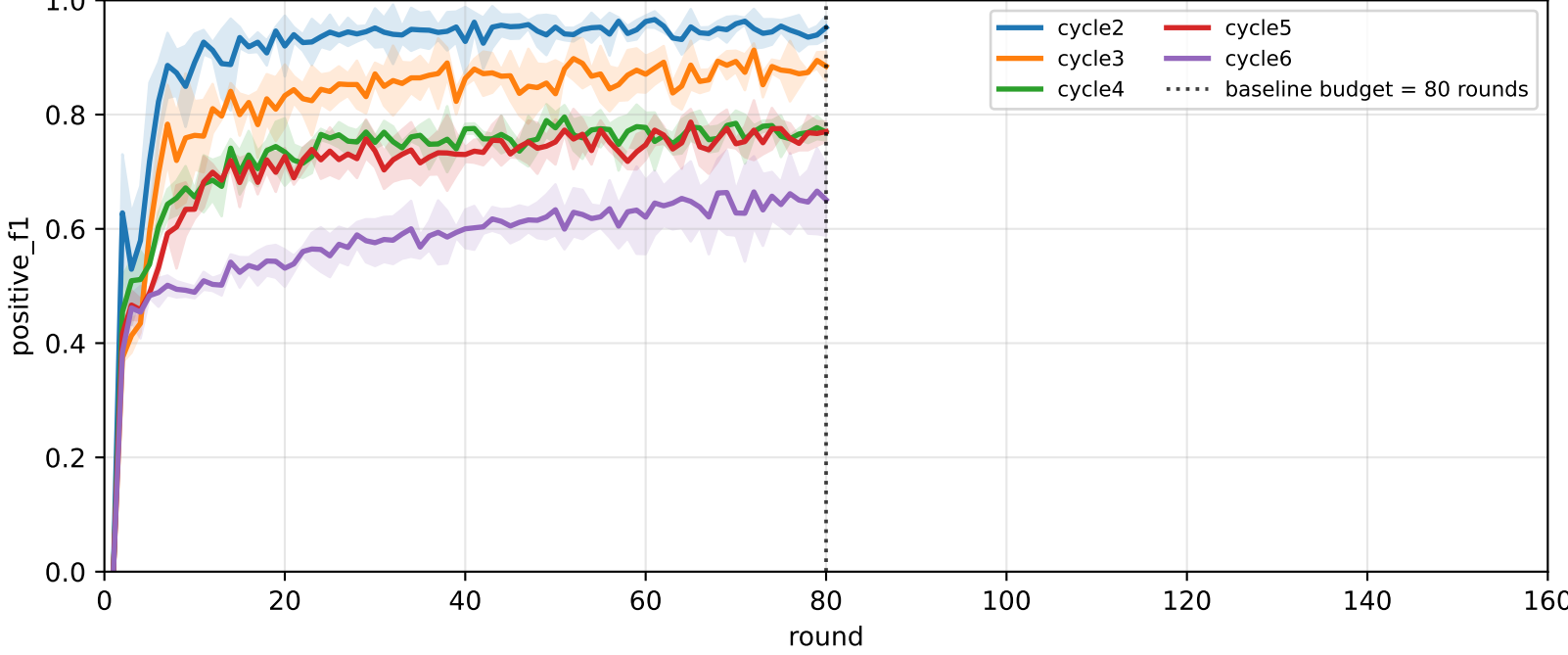
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



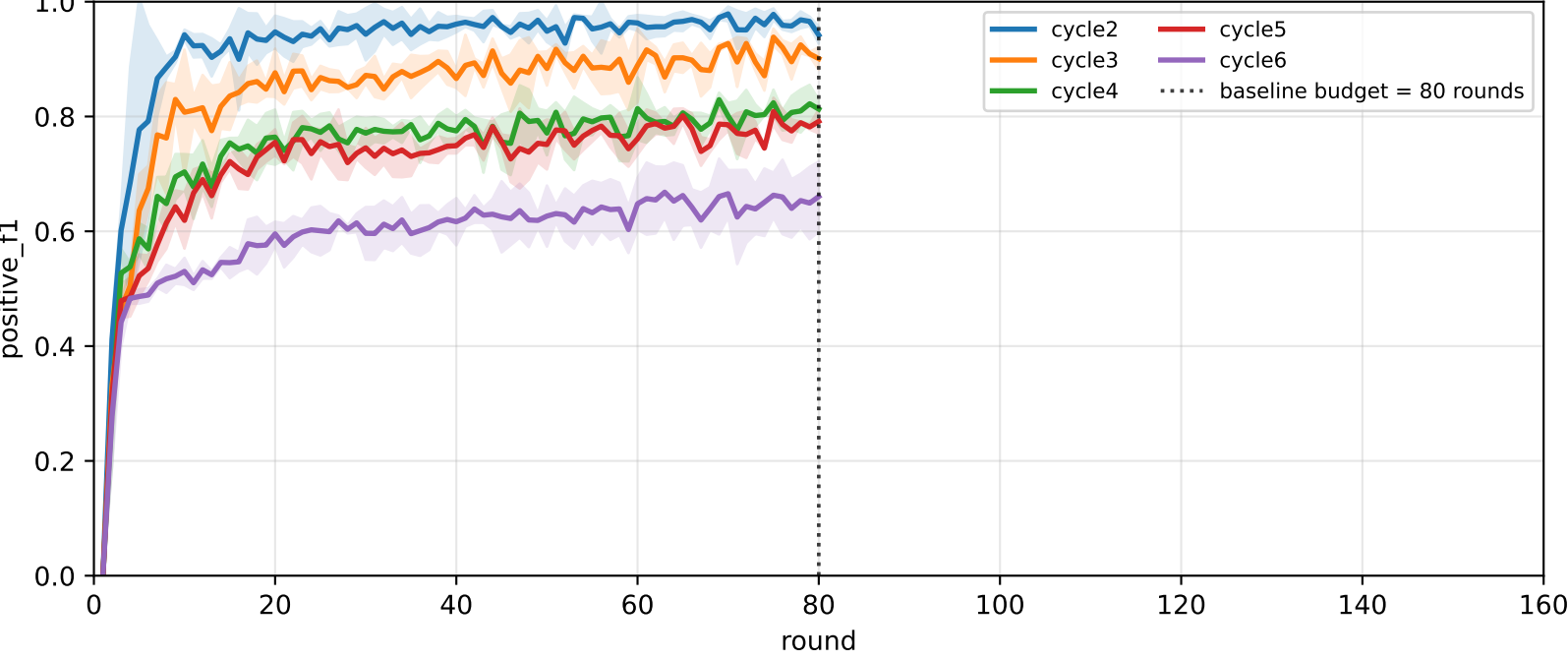
Pooled local centralized positive F1



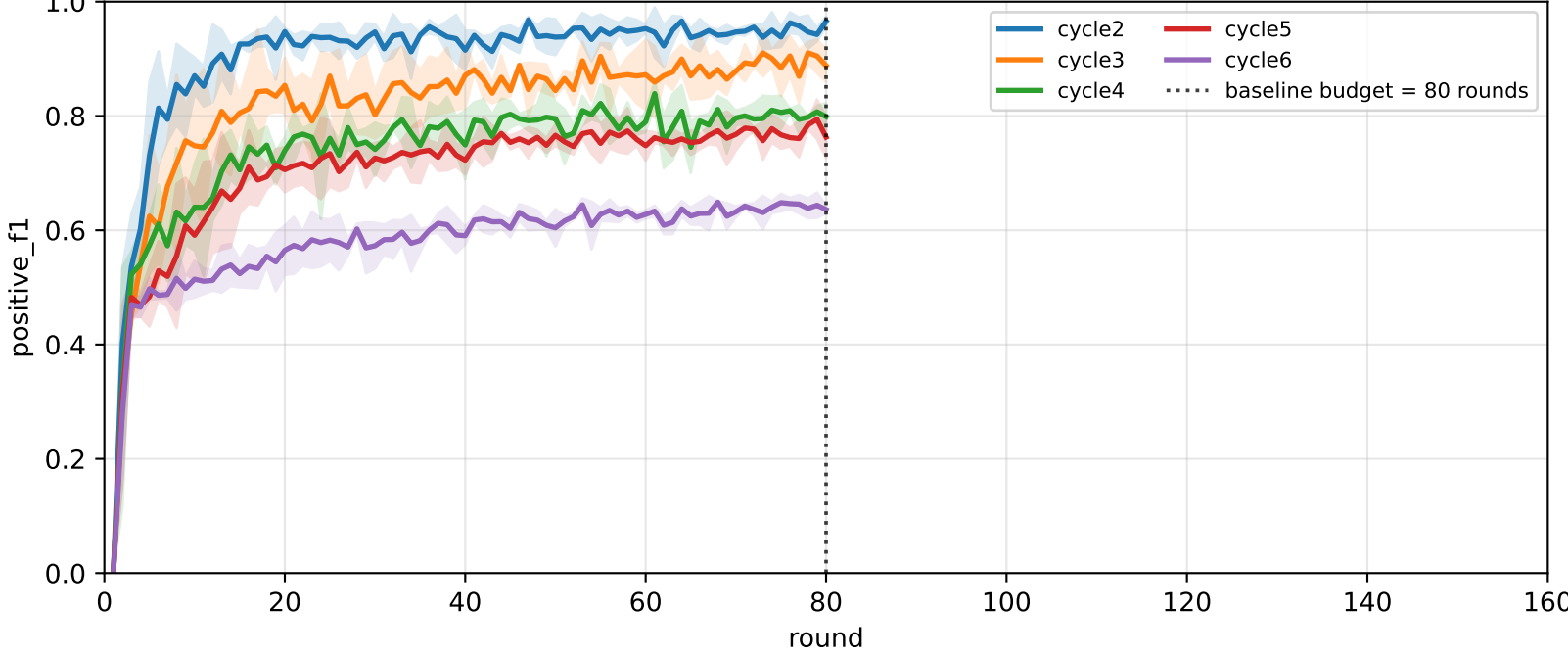
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)

